

ExonViz: Transcript visualization made easy

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Software

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9 Summary

Transcripts of a gene contain one or more **exons**, which encode the functional parts of the transcript, and **introns**, which are removed in a process called **splicing**. A single gene typically encodes multiple **transcripts** by including different exons. Protein coding genes include one or more coding exons, which encode a protein using three-letter sequences called **codons**. These codons do not necessarily coincide with exon boundaries, because a single codon can span two exons. If the codon boundaries of adjacent exons are not aligned, an often detrimental **frame shift** is introduced. Taking exon boundary reading frames, which we will call **exon boundary frames**, into account is crucial when considering the effect of mutations and when designing genetic therapies such as exon skipping.

19 ExonViz is a Python package and web application that creates biologically accurate RNA
20 transcript figures, including features such as coding regions, genetic variants and exon boundary
21 frames.

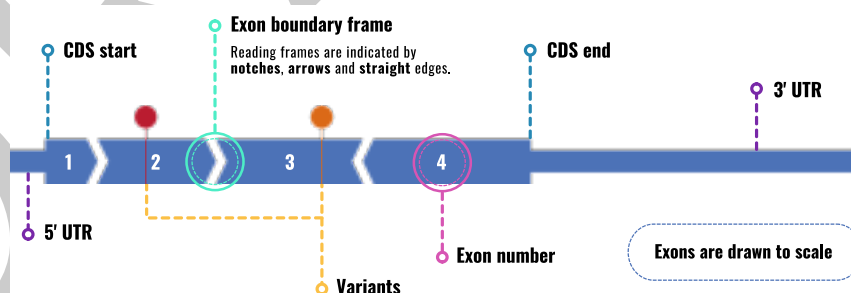


Figure 1: Example transcript highlighting ExonViz features. **5' UTR:** Non coding region at the start of the transcript. **CDS start:** Start of the coding region. **CDS end:** End of the coding region. **3' UTR:** Non coding region at the end of the transcript.

22 Statement of need

Visualization of transcripts, including features like exon boundary frames, coding and non coding regions is important within the field of clinical and human genetics (Walker et al., 2023). Illustrating the exon structure and the location of variants is common practice, especially when new genes, variants or transcripts have been discovered. These illustrations are also used to assess potential genetic treatment options (e.g., canonical exon skipping), in teaching settings, in diagnostics, to identify mutational hotspots and for genetic counseling. In particular,

29 knowledge about the exon boundary frames aids in the assessment of the pathogenicity of
30 genetic variants using the ACMG-AMP guidelines (Richards et al., 2015), when evaluating exon
31 spanning deletions (Cheerie et al., 2025) and when interpreting the effects of splice altering
32 variants (Walker et al., 2023).

33 To date, most people have to resort to manually drawing transcripts with tools like Illustrator,
34 Photoshop or BioRender, or forgo illustrations altogether. Creating transcript visualizations
35 should be quick and easy to be utilized in clinical and day to day settings, rather than to create
36 a bespoke figure for a manuscript or presentation.

37 State of the field

38 Several tools have been made available to visualize various aspects of genes and transcripts.
39 ggttranscript (Gustavsson et al., 2022) and wiggleplotr (Alasoo, 2017) can visualize transcripts
40 and exons, while tools like genepainter (Mühlhausen et al., 2015) or Swan (Reese & Mortazavi,
41 2021) can be used to visualize different transcript isoforms. Variants can be shown on the
42 transcript with Variant View (Ferstay et al., 2013). However, all of these tools require
43 substantial expertise to setup and retrieve the required transcript models, which make them
44 hard to use for users with minimal technical expertise. Furthermore, none of them have the
45 option to indicate exon boundary frames.

46 To our knowledge there are currently no easily usable tools available which allow a non-technical
47 user to quickly draw all features required for a comprehensive overview of a transcript's structure
48 and the location of variants of interest.

49 Software design

50 ExonViz is written in Python 3, its web interface is build using Flask. To avoid the complexities
51 of retrieving and processing various different transcript definitions, ExonViz uses the public
52 Mutalyzer API (Lefter et al., 2021) to fetch transcript annotations. This gives ExonViz access
53 to all transcripts defined in the RefSeq (O'Leary et al., 2016) and Ensembl (Harrison et
54 al., 2023) databases across many species, ranging from human and mouse to fruit fly and
55 coelacanth.

56 Reverse strand transcript are inverted so that ExonViz always visualizes transcripts in their
57 forward orientation. This avoids the complications that come with the inverted annotations
58 for transcripts which are annotated on the reverse strand of the chromosome. Variants are
59 assigned to their corresponding exon, which also contains the size, coding region and other
60 features which are required to draw an exon. Exons can be split to ensure they do not exceed
61 the specified page width, analogous to how long words can be split over multiple lines.

62 ExonViz can also read and write the normalized exon and variant models, allowing the user to
63 specify custom transcripts and exons in a simple TSV format.

64 Method

65 ExonViz visualizes the exon boundary frames by using different shapes for the boundary of
66 exons. Figure 2 shows all possible combinations of exon and codon boundaries, and the
67 corresponding exon boundary shapes. When the exon and codon boundaries coincide (frame
68 0) the exons are drawn with a straight edge, as is the case of exon 1 and 2. Exon 2 ends one
69 base into the codon (in frame 1), which is drawn using an arrow on the end of the exon. Exon
70 3 starts in frame 1, and is drawn with a notch at the start of the exon. This is reversed for the
71 boundary between exons 3 and 4, which is in frame 2. Since the exons of a transcript should
72 fit together, exons in conflicting frames (e.g. because of a frame shift inducing variant) are
73 easily spotted due to the fact that the exon boundaries do not fit together.



Figure 2: Visualization of the relation between codons and exon boundary frames. The shapes of the exons illustrate the relation between the exon boundaries and the codon boundaries.

The output of ExonViz is an SVG figure generated using the `svg-py` library, which can be used directly or modified using modern graphical editing programs. It is also possible to output the transcript and variants in TSV format, edit the transcript using any text editor or spreadsheet program, and draw the modified transcript using ExonViz. The [online documentation](#) has a number of examples of custom transcripts that can be visualized this way.

Research impact statement

ExonViz has proven to be a useful resource to quickly visualize exon reading frames and check the location of variants in a transcript. ExonViz is actively being used in the field of personalized medicine and is one of the recommended resources in the latest consensus guidelines for assessing pathogenic variants for RNA therapies (Cheerie et al., 2025). In addition, the [ExonViz website](#) has been used to generate over 8000 transcript figures between September 2023 and September 2025.

Conclusion

To our knowledge, ExonViz is the first publicly available application that allows for automatic visualization of transcripts with additional features such as exon boundary frames and variants along the transcript. ExonViz can be used for illustrations within publications, assessment of treatment options, teaching purposes and genetic counseling. Figures generated by ExonViz are free to use under the Creative Commons BY license. Furthermore, we allow the user to construct their own transcripts, for example to visualize non-standard exons or alternative isoforms. ExonViz can be accessed as a web application via exonviz.rnatherapy.nl or installed via [PyPI](#). The source code is available on [Github](#).

AI usage disclosure

No generative AI tools have been used for ExonViz, the documentation or this manuscript.

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